Homework 16: Matrices of Linear Transformations – Reflections and Projections

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BCS A, E, F, G

We have seen in one of our previous homework's that the standard matrix of the reflection $F_{\boldsymbol{u}}$ is

$$F_{\boldsymbol{u}} = 2\boldsymbol{u}\boldsymbol{u}^T - I\ .$$

One can compute the matrix using the unit vector $\boldsymbol{u}$ [$L = span\{\boldsymbol{u}\}$]. What if you are given with the equation of line $y = mx$ or inclination angle of that line θ.

Problem 1. ***Reflection Through The Line Given $y = mx$.***

Show that the linear transformation that reflects $\mathbb{R}^2$ through the line $y = mx$ has standard matrix

$$F_L = \frac{1}{m^2+1}\begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}.$$

Problem 2. ***Reflection Through The Line Given Inclination Angle θ.***

Show that the linear transformation that reflects $\mathbb{R}^2$ through the line that points at an angle θ counter-clockwise from the x −axis has standard matrix

$$F_\theta = \begin{bmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{bmatrix}.$$

We will work all of the scenarios regarding reflection matrices on the line $y = x$.

Problem 3. Find the standard matrices of the linear transformations that project onto the line $y = x$.

a) Using $F_{\boldsymbol{u}} = 2\boldsymbol{u}\boldsymbol{u}^T - I$: $\boldsymbol{u} = (1,0) \in \mathbb{R}^2$.

b) Using F_L: $y = x$.

c) Using F_θ: The line at an angle $\theta = \pi/4$.

Problem 4. Suppose $\boldsymbol{u} \in \mathbb{R}^n$ is a unit vector and $A = \boldsymbol{u}\boldsymbol{u}^T \in M_n$. Show that $A^2 = A$. Provide a geometric interpretation of this fact. Is it the projection matrix in $\mathbb{R}^n$?

We will try to understand this scenario in $\mathbb{R}^2$ through next problem.

Problem 5. Find the standard matrices of the linear transformations that project onto the lines in the direction of the following vectors, and depict these projections geometrically.

a) $\boldsymbol{u} = (1,0) \in\ \mathbb{R}^2$, and

b) $\boldsymbol{w} = (1,2,3) \in \mathbb{R}^3$.

Hint. Use the $A = \boldsymbol{u}\boldsymbol{u}^T$ from problem 4 to find such projections.

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Good Luck